

Ch 3 → create functions

CH-3.1

Interpolation

Find the type of function depending on the graph with different points

Lagrange polynomial:

The problem of determining a polynomial of degree one that passes through the distinct points (x_0, y_0) and (x_1, y_1) is the same as approximating a function f for which $f(x_0) = y_0$ and $f(x_1) = y_1$ by means of a first-degree polynomial **interpolating**, or agreeing with, the values of f at the given points. Using this polynomial for approximation within the interval given by the endpoints is called polynomial **interpolation**.

Define the functions

$$L_0(x) = \frac{x - x_1}{x_0 - x_1}$$

and

$$L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

★ Only when 2 points are given

$$P_1(x) = f(x_0)L_0(x) + f(x_1)L_1(x)$$

The linear **Lagrange interpolating polynomial** through (x_0, y_0) and (x_1, y_1) is

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1) = \frac{x - x_1}{x_0 - x_1}f(x_0) + \frac{x - x_0}{x_1 - x_0}f(x_1).$$

Note that

$$L_0(x_0) = 1, \quad L_0(x_1) = 0, \quad L_1(x_0) = 0, \quad \text{and} \quad L_1(x_1) = 1,$$

which implies that

$$P(x_0) = 1 \cdot f(x_0) + 0 \cdot f(x_1) = f(x_0) = y_0$$

and

$$P(x_1) = 0 \cdot f(x_0) + 1 \cdot f(x_1) = f(x_1) = y_1.$$

So P is the unique polynomial of degree at most one that passes through (x_0, y_0) and (x_1, y_1) .

Determine the linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1).

Solution In this case we have

$$L_0(x) = \frac{x-5}{2-5} = -\frac{1}{3}(x-5) \quad \text{and} \quad L_1(x) = \frac{x-2}{5-2} = \frac{1}{3}(x-2),$$

so

$$P(x) = -\frac{1}{3}(x-5) \cdot 4 + \frac{1}{3}(x-2) \cdot 1 = -\frac{4}{3}x + \frac{20}{3} + \frac{1}{3}x - \frac{2}{3} = -x + 6.$$

x_0, x_1, x_2

Degree 1

$$L_0 = \frac{x-x_1}{x_0-x_1}, \quad L_1 = \frac{x-x_0}{x_1-x_0}$$

$$p(x) = L_0(x) f(x_0) + L_1(x) f(x_1)$$

Degree 2

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}, \quad L_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}, \quad L_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

$$p(x) = L_0(x) f(x_0) + L_1(x) f(x_1) + L_2(x) f(x_2)$$

Degree 3 (max deg asked in the exam)

$$L_0(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}, \quad L_1(x) = \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}, \quad L_2(x) = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}, \quad L_3(x) = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}$$

$$L_0(x) f(x_0)$$

$$f(x) = \frac{1}{x}$$

$$x_0 = 2, \quad x_1 = 2.75, \quad x_2 = 4$$

If more than 2 points are given, to create degree 1 polynomial choose 2 points, however if degree 2 is needed choose 3 points

If $x_0, x_1, \dots, x_n$ are $n + 1$ distinct numbers and f is a function whose values are given at these numbers, then a unique polynomial $P(x)$ of degree at most n exists with

$$f(x_k) = P(x_k), \quad \text{for each } k = 0, 1, \dots, n.$$

This polynomial is given by

$$P(x) = f(x_0)L_{n,0}(x) + \cdots + f(x_n)L_{n,n}(x) = \sum_{k=0}^n f(x_k)L_{n,k}(x), \quad (3.1)$$

where, for each $k = 0, 1, \dots, n$,

$$\begin{aligned} L_{n,k}(x) &= \frac{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)} \\ &= \prod_{\substack{i=0 \\ i \neq k}}^n \frac{(x - x_i)}{(x_k - x_i)}. \end{aligned} \quad (3.2)$$

We will write $L_{n,k}(x)$ simply as $L_k(x)$ when there is no confusion as to its degree.

EXC

$$L_0(x) = \frac{(x-2.75)(x-4)}{(2-2.75)(2-4)} = \frac{(x-2.75)(x-4)}{1.5}$$

$$L_1(x) = \frac{(x-2)(x-4)}{(2.75-2)(2.75-4)} = \frac{(x-2)(x-4)}{-0.9375}$$

$$L_2(x) = \frac{(x-2)(x-2.75)}{(4-2)(4-2.75)} = \frac{(x-2)(x-2.75)}{2.5}$$

$$p(x) = \frac{1}{2} \left[\frac{(x-2.75)(x-4)}{1.5} \right] + \frac{1}{2.75} \left[\frac{(x-2)(x-4)}{-0.9375} \right] + \frac{1}{4} \left[\frac{(x-2)(x-2.75)}{2.5} \right] = 0.32955$$

about $\frac{1}{3} = 0.3333$

- (a) Use the numbers (called *nodes*) $x_0 = 2$, $x_1 = 2.75$, and $x_2 = 4$ to find the second Lagrange interpolating polynomial for $f(x) = \Gamma/x$. $f(x_0) = \frac{1}{2}$ $f(x_1) = \frac{1}{2.75}$ $f(x_2) = \frac{1}{4}$
- (b) Use this polynomial to approximate $f(3) = 1/3$.

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-2.75)(x-4)}{(2-2.75)(2-4)}$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-2)(x-4)}{(2.75-2)(2.75-4)}$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-2)(x-2.75)}{(4-2)(4-2.75)}$$



Solution (a) We first determine the coefficient polynomials $L_0(x)$, $L_1(x)$, and $L_2(x)$. In nested form they are

$$L_0(x) = \frac{(x - 2.75)(x - 4)}{(2 - 2.5)(2 - 4)} = \frac{2}{3}(x - 2.75)(x - 4),$$

$$L_1(x) = \frac{(x - 2)(x - 4)}{(2.75 - 2)(2.75 - 4)} = -\frac{16}{15}(x - 2)(x - 4),$$

and

$$L_2(x) = \frac{(x - 2)(x - 2.75)}{(4 - 2)(4 - 2.5)} = \frac{2}{5}(x - 2)(x - 2.75).$$

Also, $f(x_0) = f(2) = 1/2$, $f(x_1) = f(2.75) = 4/11$, and $f(x_2) = f(4) = 1/4$, so

$$\begin{aligned} P(x) &= \sum_{k=0}^2 f(x_k)L_k(x) \\ &= \frac{1}{3}(x - 2.75)(x - 4) - \frac{64}{165}(x - 2)(x - 4) + \frac{1}{10}(x - 2)(x - 2.75) \\ &= \frac{1}{22}x^2 - \frac{35}{88}x + \frac{49}{44}. \end{aligned}$$

(b) An approximation to $f(3) = 1/3$ (see Figure 3.6) is

$$f(3) \approx P(3) = \frac{9}{22} - \frac{105}{88} + \frac{49}{44} = \frac{29}{88} \approx 0.32955.$$

Recall that in the opening section of this chapter (see Table 3.1) we found that no Taylor polynomial expanded about $x_0 = 1$ could be used to reasonably approximate $f(x) = 1/x$ at $x = 3$. ■

Exercise Set 3.1

1. For the given functions $f(x)$, let $x_0 = 0$, $x_1 = 0.6$, and $x_2 = 0.9$. Construct interpolation polynomials of degree at most one and at most two to approximate $f(0.45)$, and find the absolute error.

a. $f(x) = \cos x$

b. $f(x) = \sqrt{1+x}$

c. $f(x) = \ln(x+1)$ $L_0 = \frac{x-x_1}{x_0-x_1} = \frac{0.45-0.6}{0-0.6} = \frac{1}{4}$

d. $f(x) = \tan x$ $L_1 = \frac{x-x_0}{x_1-x_0} = \frac{0.45-0}{0.6-0} = \frac{3}{4}$

$$L_0 = \frac{(x-x_1)}{(x_0-x_1)} = \frac{(0.45-0.6)}{0-0.6} = \frac{1}{4}$$

$$L_1 = \frac{(x-x_0)}{(x_1-x_0)} = \frac{(0.45-0)}{0.6-0} = \frac{3}{4}$$

$$\frac{1}{4} \ln(0+1) + \frac{3}{4} \ln(0.6+1) \\ = 0.352503$$

$$p(x) =$$

1. For the given functions $f(x)$, let $x_0 = 0$, $x_1 = 0.6$, and $x_2 = 0.9$. Construct interpolation polynomials of degree at most one and at most two to approximate $f(0.45)$, and find the absolute error.

a. $f(x) = \cos x$

c. $f(x) = \ln(x+1)$

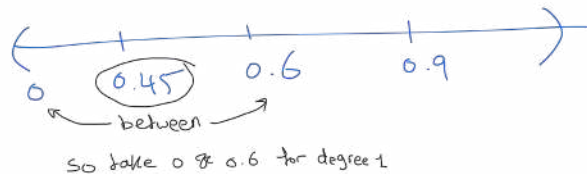
Deg 1: $x_0 = 0, x_1 = 0.6$

$$f(x_0) = \cos 0 = 1$$

$$L_0(x) = \frac{x - 0.6}{0 - 0.6} = \frac{x - 0.6}{-0.6}$$

$$f(x_1) = \cos(0.6)$$

$$L_1(x) = \frac{x - 0}{0.6 - 0} = \frac{x}{0.6}$$



$$P(x) = \frac{x - 0.6}{-0.6} (1) + \frac{x}{0.6} \cos(0.6)$$

$$P(0.45) = \frac{0.45 - 0.6}{-0.6} + \frac{0.45}{0.6} \cos(0.6) = 0.869002$$

$$\text{Exact value: } \cos(0.45) = 0.90045$$

$$\text{Absolute error} = |0.90045 - 0.869002| = 0.031448$$

The more the degree is, the difference in absolute error is less.

Deg 2:

$$L_0(x) = \frac{(x - 0.6)(x - 0.9)}{(0 - 0.6)(0 - 0.9)}$$

$$L_1(x) = \frac{(x - 0)(x - 0.9)}{(0.6 - 0)(0.6 - 0.9)}$$

$$L_2(x) = \frac{(x - 0)(x - 0.6)}{(0.9 - 0)(0.9 - 0.6)}$$

$$P(x) = \frac{(x - 0.6)(x - 0.9)}{0.54} (1) + \frac{(x - 0.9)}{-0.18} (\cos(0.6)) + \frac{x(x - 0.6)}{0.27} \cos(0.9)$$

$$P(x) = \frac{(0.45 - 0.6)(0.45 - 0.9)}{0.54} (1) + \frac{(0.45 - 0.9)}{-0.18} (\cos(0.6)) + \frac{0.45(0.45 - 0.6)}{0.27} \cos(0.9) = 0.8981$$

$$\text{Exact value: } \cos(0.45) = 0.90045$$

$$\text{Absolute error} = |0.90045 - 0.8981| = 0.0023$$

Exercise

create/find the polynomial first

then find the value

5. Use appropriate **Lagrange interpolating polynomials** of degrees one, two, and three **to approximate** each of the following:
- a. $f(8.4)$ if $f(8.1) = 16.94410$, $f(8.3) = 17.56492$, $f(8.6) = 18.50515$, $f(8.7) = 18.82091$
 - b. $f(-\frac{1}{3})$ if $f(-0.75) = -0.07181250$, $f(-0.5) = -0.02475000$, $f(-0.25) = 0.33493750$, $f(0) = 1.10100000$
 - c. $f(0.25)$ if $f(0.1) = 0.62049958$, $f(0.2) = -0.28398668$, $f(0.3) = 0.00660095$, $f(0.4) = 0.24842440$
 - d. $f(0.9)$ if $f(0.6) = -0.17694460$, $f(0.7) = 0.01375227$, $f(0.8) = 0.22363362$, $f(1.0) = 0.65809197$

5. Use appropriate Lagrange interpolating polynomials of degrees one, two, and three to approximate each of the following:

b. $f(-\frac{1}{3})$ if $f(-0.75) = -0.07181250$, $f(-0.5) = -0.02475000$, $f(-0.25) = 0.33493750$, $f(0) = 1.10100000$

$$-0.25 < f(-\frac{1}{3}) < -0.5$$

Deg 1° $x_0 = -0.5$ $x_1 = -0.25$

$$f(-0.5) = -0.02475$$

$$f(-0.25) = 0.3349375$$

$$L_0(x) = \frac{(x+0.25)}{-0.5-(-0.25)} = \frac{(x+0.25)}{-0.25}$$

$$L_1(x) = \frac{(x+0.5)}{-0.25+0.5} = \frac{(x+0.5)}{0.25}$$

$$p(x) = L_0(x)f(x_0) + L_1(x)f(x_1) = \frac{x+0.25}{-0.25}(-0.02475) + \frac{x+0.5}{0.25}(0.3349375)$$

$$p(-\frac{1}{3}) = L_0(x)f(x_0) + L_1(x)f(x_1) = \frac{-\frac{1}{3}+0.25}{-0.25}(-0.02475) + \frac{-\frac{1}{3}+0.5}{0.25}(0.3349375) = 0.2$$

Deg 2° $x_0 = -0.75$ $x_1 = -0.5$ $x_2 = -0.25$ (u can choose the first 3 values or the next 3 values, both would be correct)

★ If u want, choose the points that has a zero as it'll be easier

$$f(-0.75) = -0.0718125$$

$$f(-0.5) = -0.02475$$

$$f(-0.25) = 0.3349375$$

$$L_0(x) = \frac{(x+0.5)(x+0.25)}{(-0.75+0.5)(-0.75+0.25)}$$

$$L_1(x) = \frac{(x+0.75)(x+0.25)}{(-0.5+0.75)(-0.5+0.25)}$$

$$L_2(x) = \frac{(x+0.75)(x+0.5)}{(-0.25+0.75)(-0.25+0.5)}$$

$$p(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2)$$

$$p(-\frac{1}{3}) = \frac{(-\frac{1}{3}+0.5)(-\frac{1}{3}+0.25)}{(-0.75+0.5)(-0.75+0.25)}(-0.0718125) + \frac{(-\frac{1}{3}+0.75)(-\frac{1}{3}+0.25)}{(-0.5+0.75)(-0.5+0.25)}(-0.02475) + \frac{(-\frac{1}{3}+0.75)(-\frac{1}{3}+0.5)}{(-0.25+0.75)(-0.25+0.5)}(0.3349375)$$

$$= 0.00797917 - 0.01375 + 0.1860763889$$

$$= 0.180303556$$

5. Use appropriate Lagrange interpolating polynomials of degrees one, two, and three to approximate each of the following:

b. $f(-\frac{1}{3})$ if $f(-0.75) = -0.07181250$, $f(-0.5) = -0.02475000$, $f(-0.25) = 0.33493750$, $f(0) = 1.10100000$

Deg 3 $x_0 = -0.75, x_1 = -0.5, x_2 = -0.25, x_3 = 0$

$$f(-0.75) = -0.0718125$$

$$f(-0.5) = -0.02475$$

$$f(-0.25) = 0.3349375$$

$$f(0) = 1.101$$

$$L_0(x) = \frac{(x+0.5)(x+0.25)(x-0)}{(-0.75+0.5)(-0.75+0.25)(-0.75-0)}$$

$$L_1(x) = \frac{(x+0.75)(x+0.25)(x-0)}{(-0.5+0.75)(-0.5+0.25)(-0.5-0)}$$

$$L_2(x) = \frac{(x+0.75)(x+0.5)(x-0)}{(-0.25+0.75)(-0.25+0.5)(-0.25-0)}$$

$$L_3(x) = \frac{(x+0.75)(x+0.5)(x+0.25)}{(0+0.75)(0+0.5)(0+0.25)}$$

$$p(-\frac{1}{3}) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2) + L_3(x)f(x_3)$$

$$= \frac{(-\frac{1}{3}+0.5)(-\frac{1}{3}+0.25)(-\frac{1}{3}-0)}{(-0.75+0.5)(-0.75+0.25)(-0.75-0)} (-0.0718125) + \frac{(-\frac{1}{3}+0.75)(-\frac{1}{3}+0.25)(-\frac{1}{3}-0)}{(-0.5+0.75)(-0.5+0.25)(-0.5-0)} (-0.02475)$$

$$+ \frac{(-\frac{1}{3}+0.75)(-\frac{1}{3}+0.5)(-\frac{1}{3}-0)}{(-0.25+0.75)(-0.25+0.5)(-0.25-0)} (0.3349375) + \frac{(-\frac{1}{3}+0.75)(-\frac{1}{3}+0.5)(-\frac{1}{3}+0.25)}{(0+0.75)(0+0.5)(0+0.25)} (1.101)$$

$$= 0.17451852$$